



SCIENCE ORBIT

start is endless



Q1

Nuclear envelope

- * double membrane
 - inner
 - outer
- * perforated - nuclear pores
- regulate entry & exit of substances

Nucleus

Nuclear matrix

- * made of protein filaments
- * extended throughout the interior
- * chromatin embedded

chromatin

- * appear as diffused mass - electron micrographs (non-dividing cells)
- * complex
 - DNA
 - proteins
- * Nuclear divisions

- chromatin condenses
- coil tightly
- forming threads
- chromosomes

constant in species
(46) - human

Nucleolus

- * appear as darkly stained granule
- * fibres adjoining part of nucleus
- chromatin

Nuclear lamina

- * made of protein filaments
- * line the interior side

Q2 Appearance of 18th living cells
- evidence from experiments & observations

- chemistry
- geology
- physics

Q3 characteristics of organisms

- development - irreversible change occur during lifespan of an organism
- Evolution - ability of organisms to change over time as a result of genetic modification
- metabolism - sum of all chemical activities taking place in an organism
- growth - irreversible increase in dry mass characterized only by the living
- irritability & coordination → movement occur in organisms.

04

No: _____

Date: ____/____/____

RNA

mRNA
(messenger)

tRNA
(transfer)

rRNA
(ribosomal)

* linear
* least abundant

* smallest
* linear-shaped
* three-looped structure

* most abundant
* complex irregular structure

↳ copy genetic info stored in DNA as a sequence of nitrogenous bases

↳ transport AA to protein synthesis site

↳ provide site where polypeptide chain assembled.

↳ transport genetic info from nucleoplasm to protein synthesis site thru nucleopores

protein synthesis site thru nucleopores

05 Plasma membrane - Functions (proteins)

↳ act as receptor molecules — Biochemicals

↳ hormones
↳ neurotransmitter
↳ immune proteins

↳ act as enzymes

↳ microvilli of epithelial lining of cells of some parts of gut — digestive enzymes in cell membrane

↳ maintain cell shape

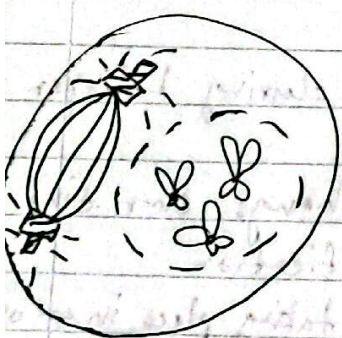
↳ attachment to cytoskeletal fibers

↳ cell recognition

(enable nearby cells to communicate with each other)

06 Mitosis

Prophase



1) Chromatin

↳ condense → shorten & thicken
↳ transform into chromosomes

2) nucleoli disappear

3) mitotic spindle formation

accumulated microtubule complex

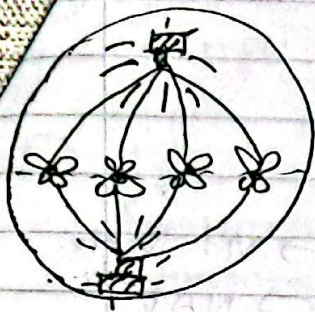
↳ centrosome

↳ aster

↳ spindle microtubule

4) microtubule lengthen } centrosome move to opp poles

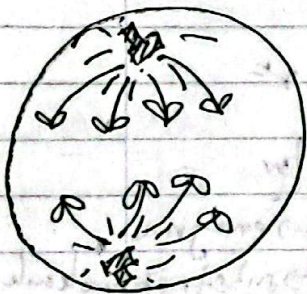
↳ visible thru microscope
↳ appear with 2 sister chromatid attached at centromere
↳ chromosomal arms attached - cohesin



metaphase

- 1) centrosomes reach opposite poles
 - 2) chromosomes — arrive @ metaphase
 - 3) centromeres
all chromosomes
↓
located @ metaphase plate
- located @ metaphase plate

End → each chromosome attached to kinetochore
microtubule @ centromere
to align @ metaphase plate.



Anaphase

- 1) sister chromatids separated @ centromere
- 2) microtubules attached to kinetochore
↳ shortens
↳ pulls sister chromatids toward opp poles
- 3) non-kinetochore microtubules lengthen

End → equal & complete set of chromosomes
found @ each pole of the cell.

Q7 Enzymes

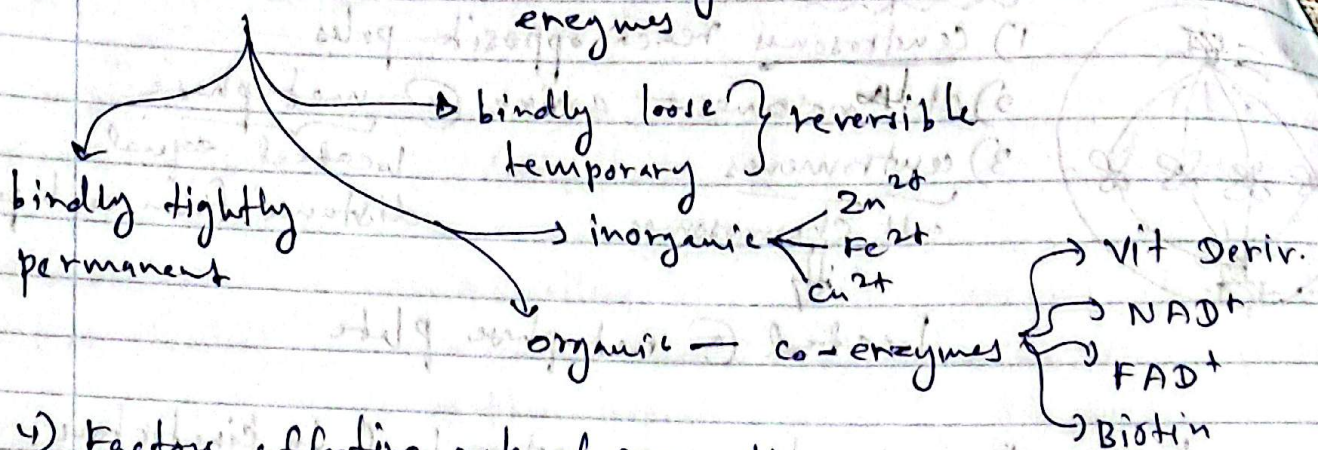
- 1) active site — formed by only few AA
others — maintain cell shape.

- a) Regulatory molecules
 - activators
 - inhibitors
 bind to specific regulatory sites
(except active site)

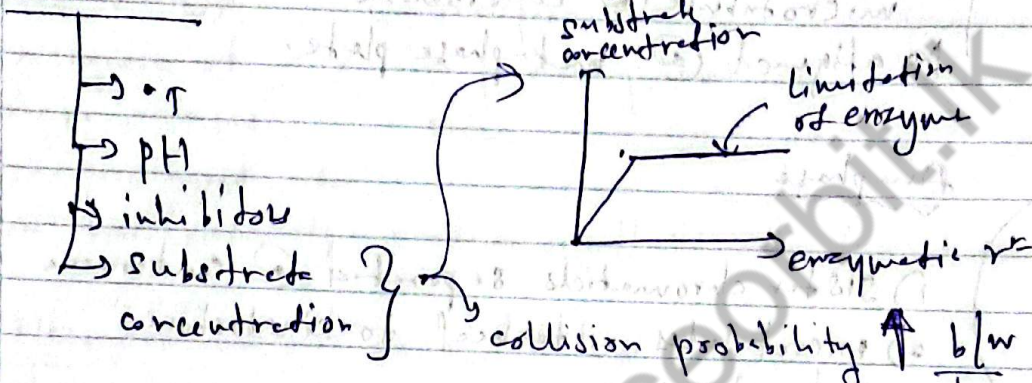
non-covalent interactions

3) Cofactors

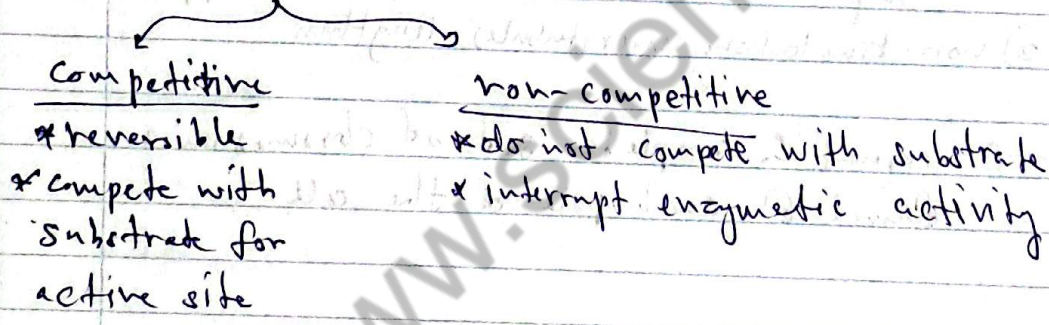
non-proteinous components essential for catalytic activities of certain enzymes



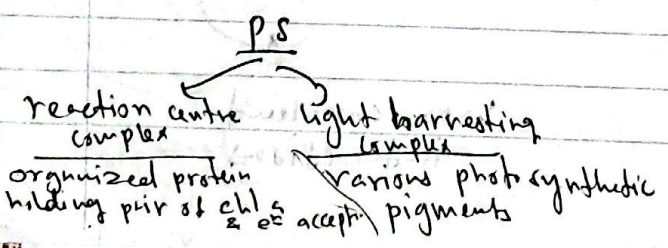
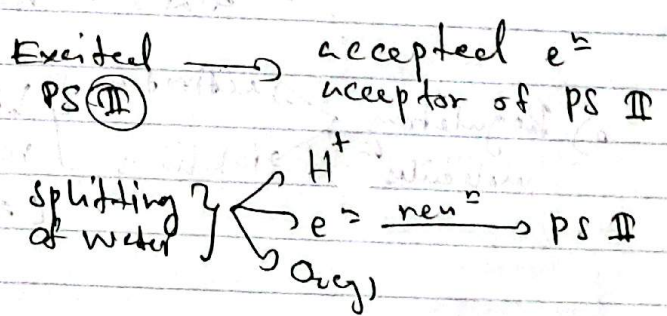
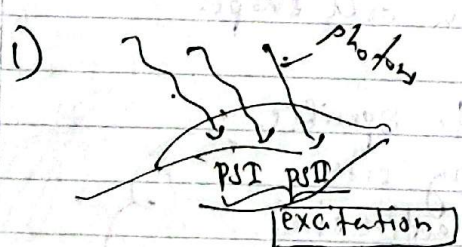
4) Factors affecting rate of enzymatic reaction



5) Inhibitors



⑥ Photosynthesis - Linear electron flow



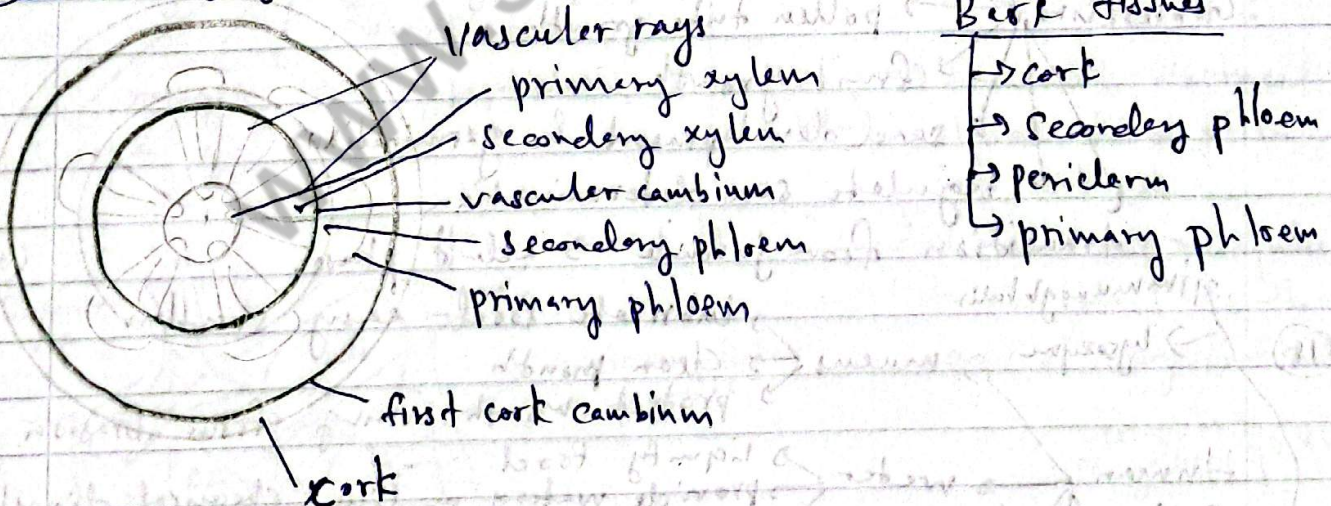
- No: _____ Date: _____
- Artificial groups
- protists — different evolutionary origins (polyphyletic group)
 - virus — not cellular organization

phyla	Bryophyte	Lyc.	Ptero	Gymno	Cycado	Angio
homolporous	sporo		sporo			
heteroporous		sporo			sporo	
dominant	gameto	sporo	sporo		sporo	
unisexual	✓					
bisexual		✓	✓		✓	

- (11) Dioecious animals
- Nematocha
 - Arthropoda
 - Echinodermata

- (12) Wood
- soft wood → gymnosperms → xylem vessels absent
 - hard wood → angiosperms → don't transport water & minerals (secondary xylem)
 - heart wood → protect from fungi

(13) Secondary growth



- (14) Phloem translocation
- mesophyll cells → phloem → Active transport (ATP) ✓
 - sugar transported

(15)

Symbiosis

mutualism

parasitism

commensalism

* coralloid root of
cycas with Anabaena

* Loranthus &
host (hemiparasitic)

* epiphytic orchids

* mycorrhizae

* Cuscuta (Dodder) & host

* Rhizobium

(16)

photo-synthetic
non-photo-synthetic

Brya
Pogo

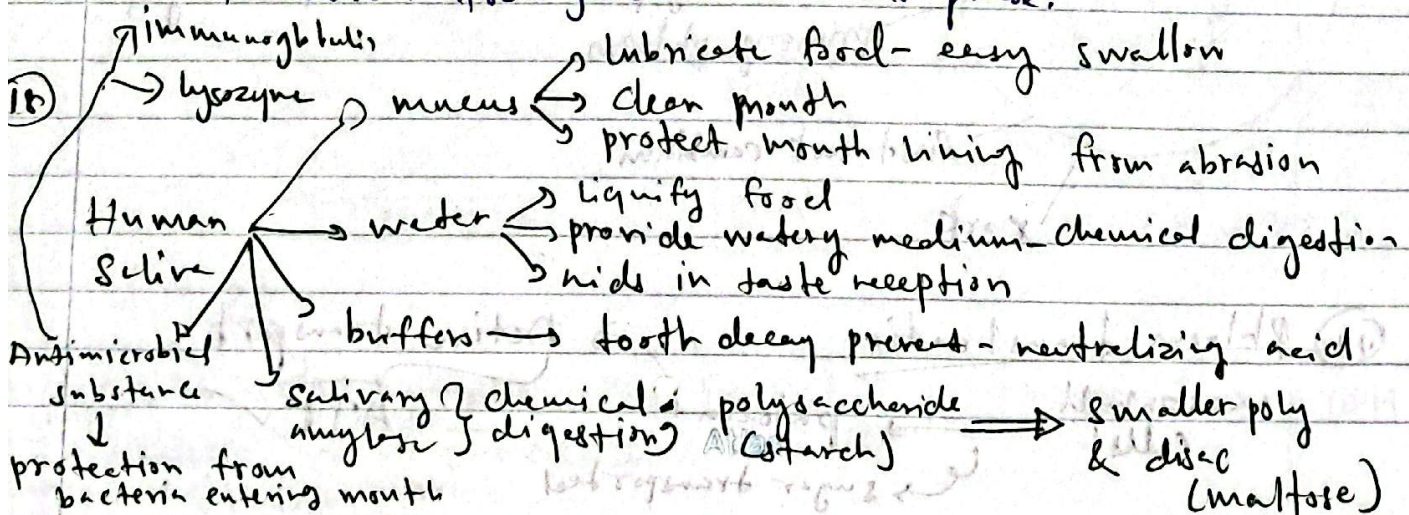
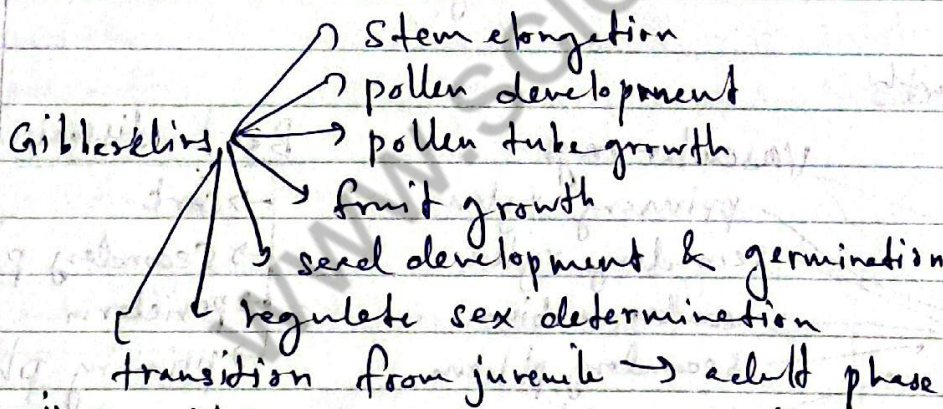
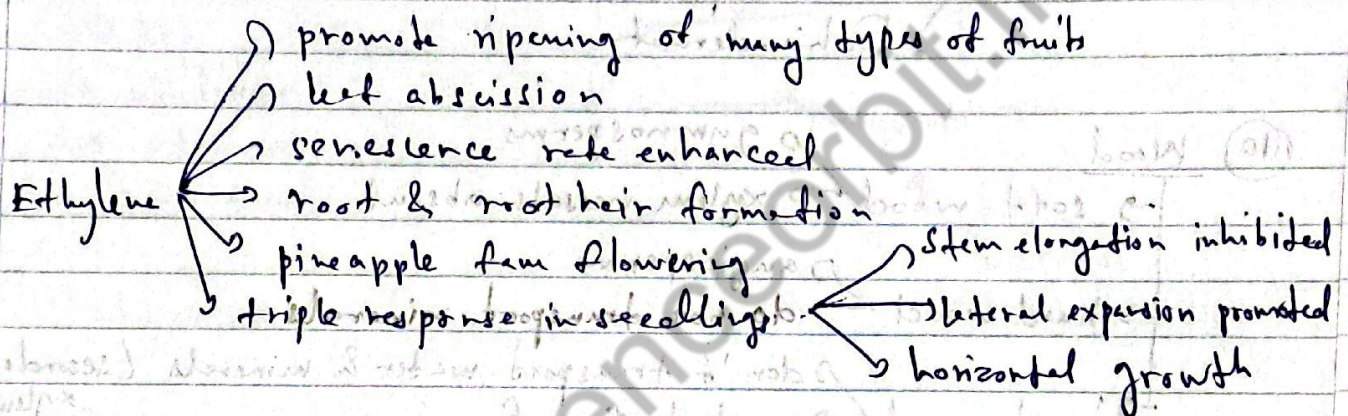
Lycor
Selagi

Plern
Nephro

Cycado
Lycor

Anthr
Coco

(17)



carb digestionpolysac $\xrightarrow{\text{pancreatic amylase}}$ DisaccharideDisac $\xrightarrow{\text{Disaccharidase}}$ monosacprotein digestion

small polypept

 $\xrightarrow{\text{pancreatic trypsin / chymotrypsin}}$

smaller polypept.

smaller polypept.

 $\xrightarrow{\text{pancreatic carboxypeptidase}}$

small pept + AA

small pept

 $\xrightarrow{\text{dipeptidase}}$

AA

 $\xrightarrow{\text{carboxypeptidase, amino peptidase}}$ fat digestion

fats

 $\xrightarrow{\text{bile salts}}$

emulsified

triglycerides

 $\xrightarrow{\text{pancreatic lipase}}$

glycerol / f.a. / monoglycerides

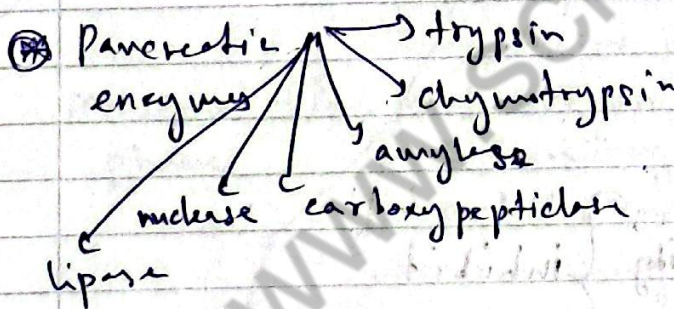
nucleic acid digestion

DNA/RNA

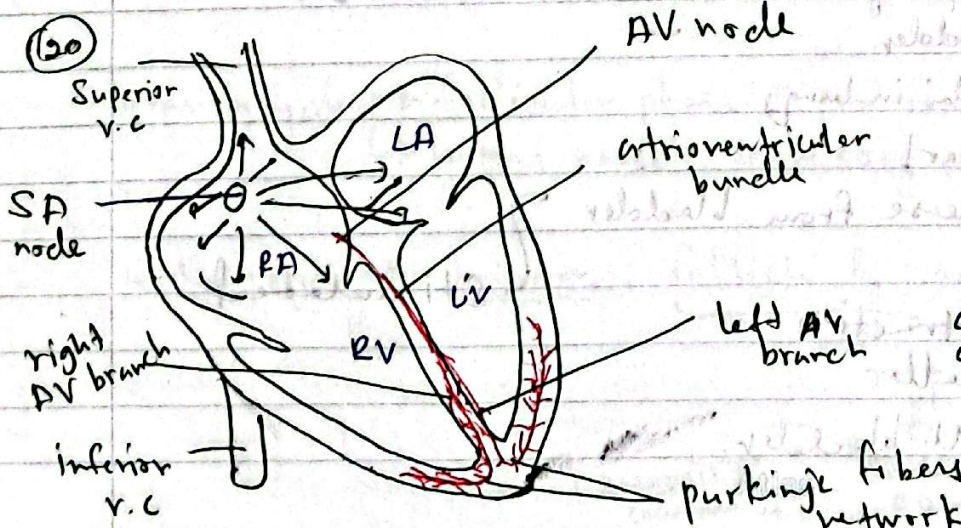
 $\xrightarrow{\text{pancreatic nuclease}}$

nucleotide

nucleotide

 $\xrightarrow{\text{nucleotidase}}$ N-base / pentose sugar / PO_4^{3-} $\xrightarrow{\text{nucleosidase, phosphatases}}$ SA node $\rightarrow$ myocardium (R.A) - superior v.c.AV node $\rightarrow$ transmit electrical signals from atria into ventricles

electrical signal impulse transmitted to the apex of myocardium



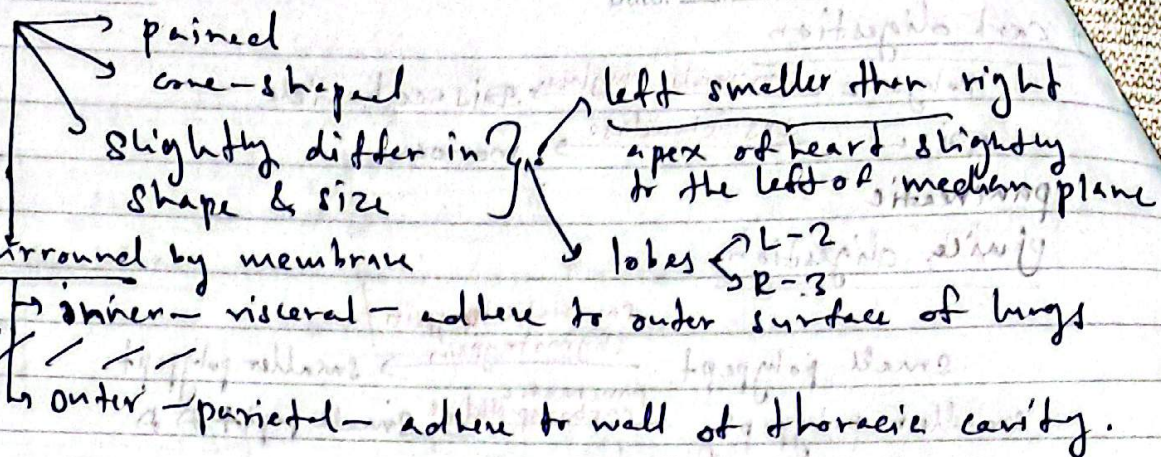
complete cardiac cycle

- complete cardiac cycle $\rightarrow$ 0.4s
- Atrial systole $\rightarrow$ 0.1s
- ventricular systole $\rightarrow$ 0.3s

(20)

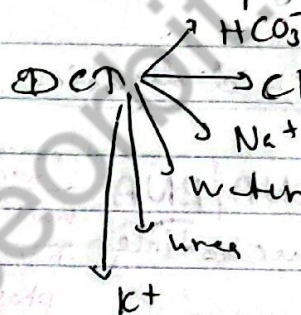
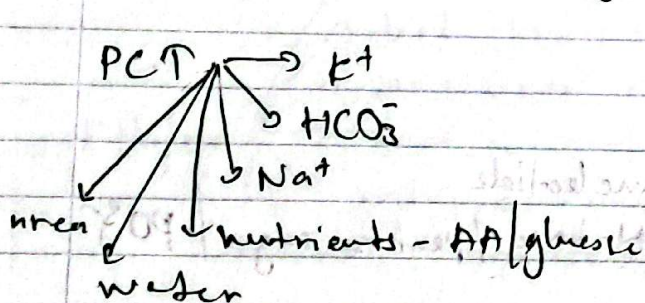
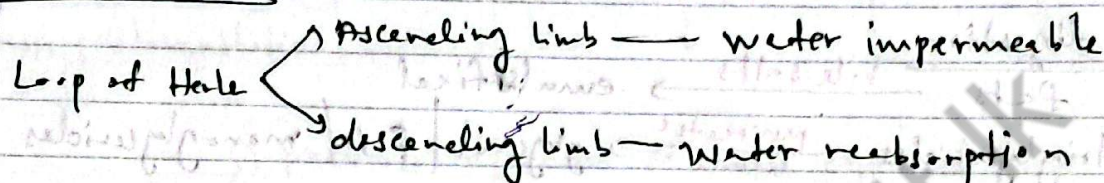
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Lungs



(21)

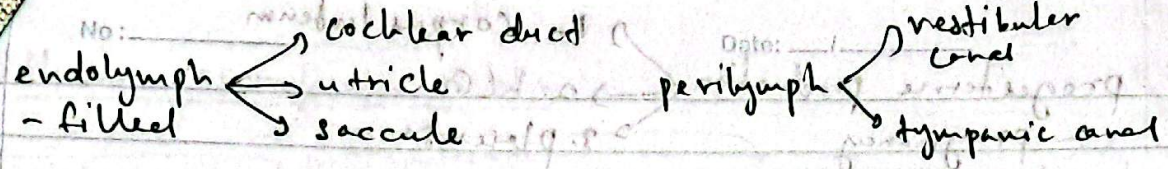
Urine Formation



(22)

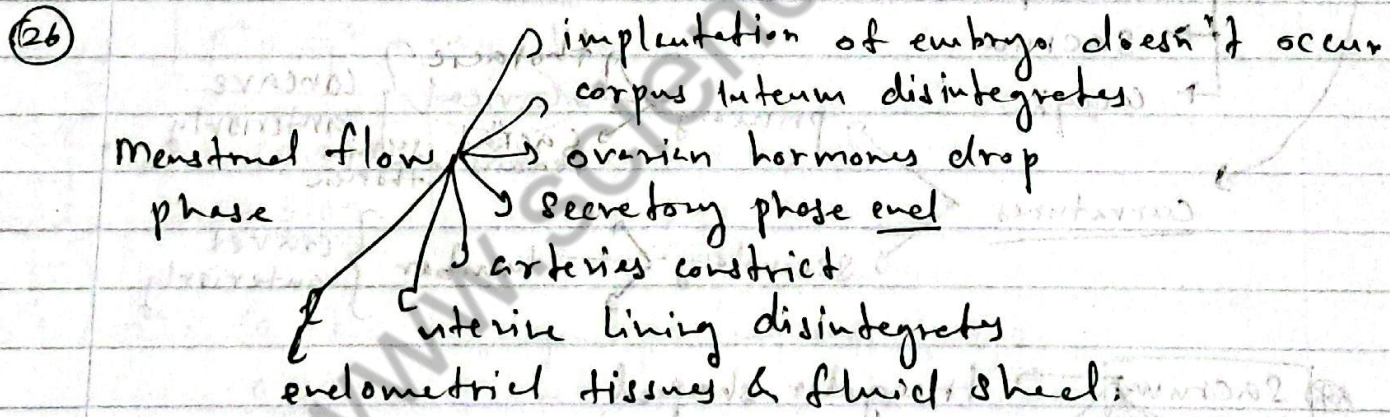
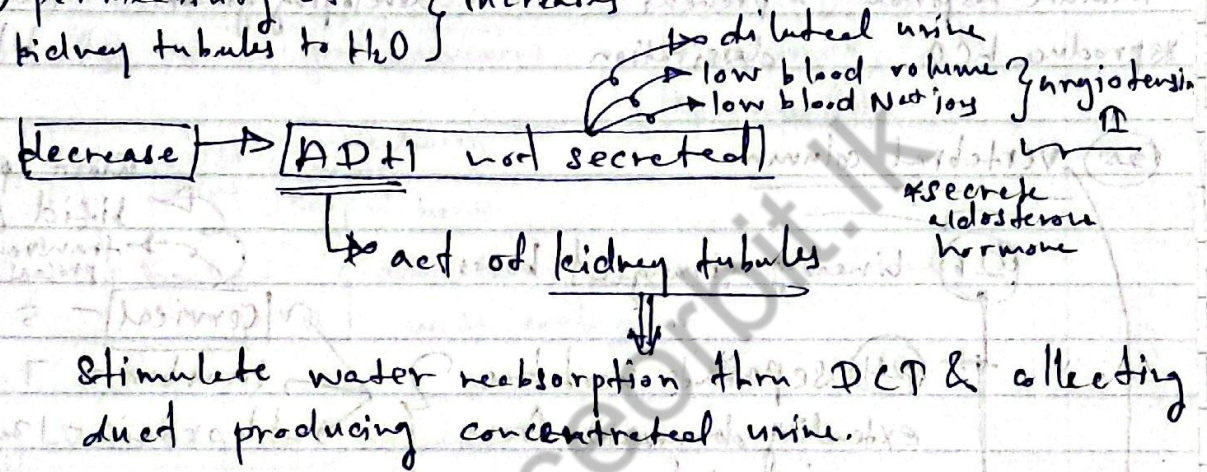
Sympathetic division

- dilate pupil of the eye
 - salivary gland secretion
 - stomach & intestine activity
 - pancreas activity
 - emptying bladder
 - relax bronchi in lungs
 - accelerate heart
 - glucose release from bladder
 - ejaculation
 - vaginal contraction
 - adrenal medulla
 - inhibit gall bladder.
- inhibit
- stimulated



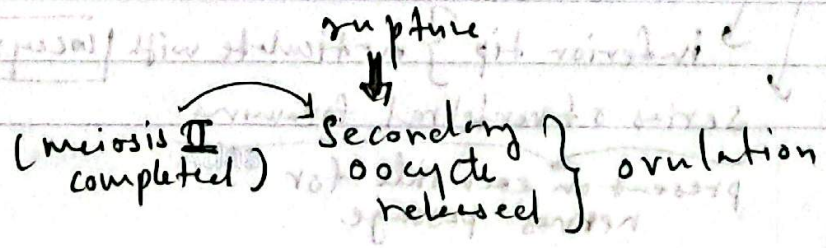
(25) blood osmolality increased beyond physiological limits;

- 1) ADH release } increases
by post. pituitary
- 2) amount of urine } decreased } concentrated
produced
- 3) permeability of } increases
kidney tubules to H_2O



ovarian cycle $\rightarrow$ follicular phase = proliferative phase & uterine cycle
 $\rightarrow$ luteal phase = secretory phase

LH surge $\Rightarrow$ triggers; follicle & adjacent ovary wall

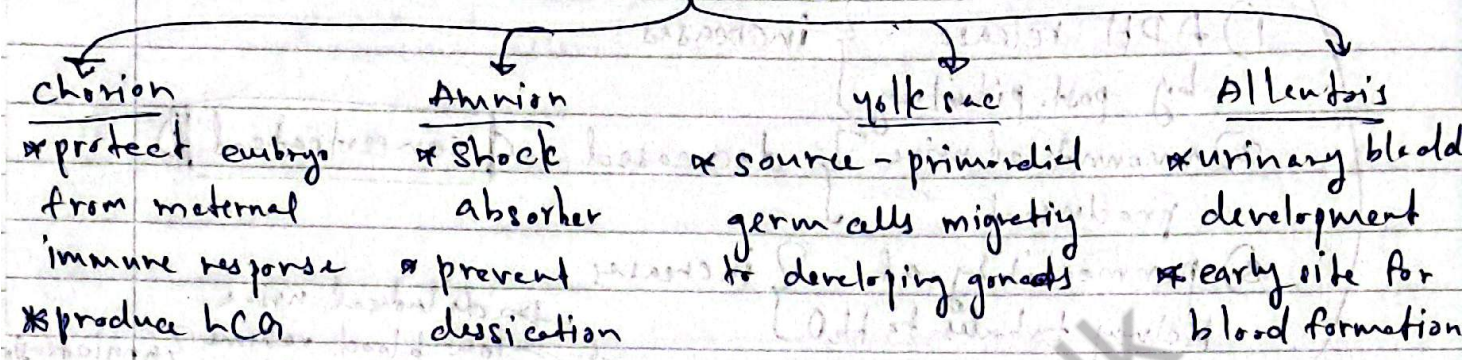


progesterone production - pregnancy

- 1. Corpus luteum
- 2. hCG
- 3. placenta

(27)

Embryonic membranes - functions



(28)

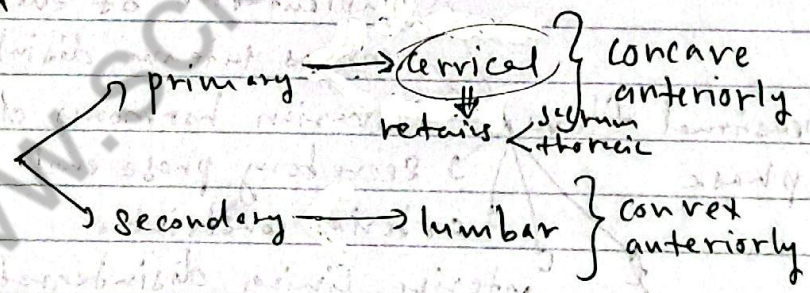
Vertebral column

(26) linearly arranged bones

(24) separate vertebrae extending downwards from occipital bone

sacrum
coccyx

Curvatures



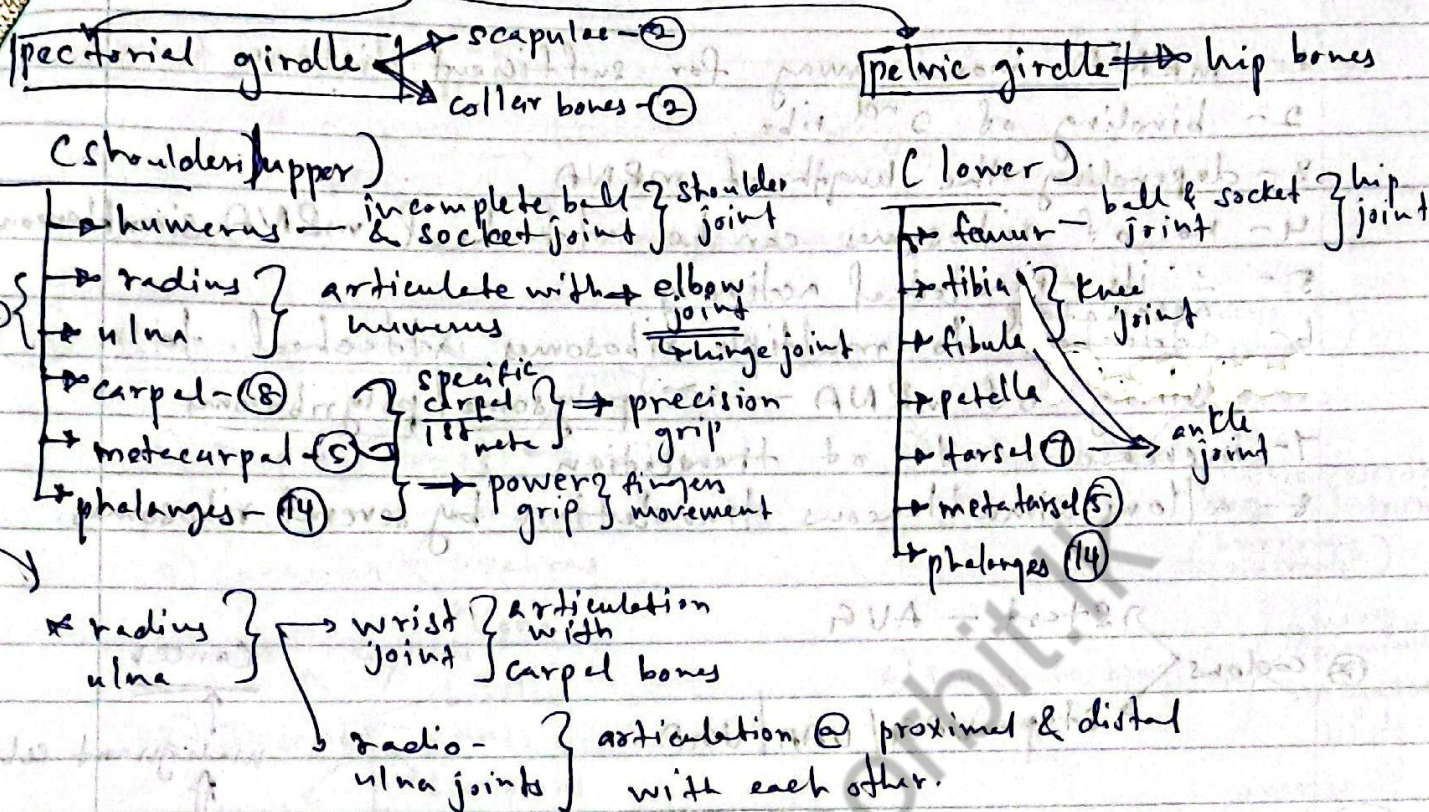
most spinous process bifid [C6-L6].
transverse process
foramen each side
vertebral artery passage.

Sacrum

triangular shaped
large bone
(5) fused rudimentary vertebrae
concave anterior surface
upper part; articulate with 5th lumbar vertebra
each side } articulate with pelvic girdle
inferior tip } articulate with coccyx
series of vertebral foramina present on each side for nerves passage.

(4) fused
terminal vertebral
small
triangular bone
broad base } articulate @ sacrum tip

Human appendicular skeleton



(30) Hardy-Weinberg eqⁿ; equation

$$p^2 + q^2 + 2pq = 1$$

gene type frequency for homozygous

$$q^2 = (0.3)^2 = 0.09$$

$$pq = 0.7 \times 0.3 = 0.21$$

$$p^2 = (0.7)^2 = 0.49$$

∴ genotype frequency } 0.21
heterozygous plants }

- (31) polyploidy - plants
- large organs
 - lower growth rates
 - flowering later / over long period of time
 - reduced fertility
 - seedless varieties production
- roots, stems, fruits, leaves, seeds, tubers, flowers

(32) Polysomes (polyribosomes)

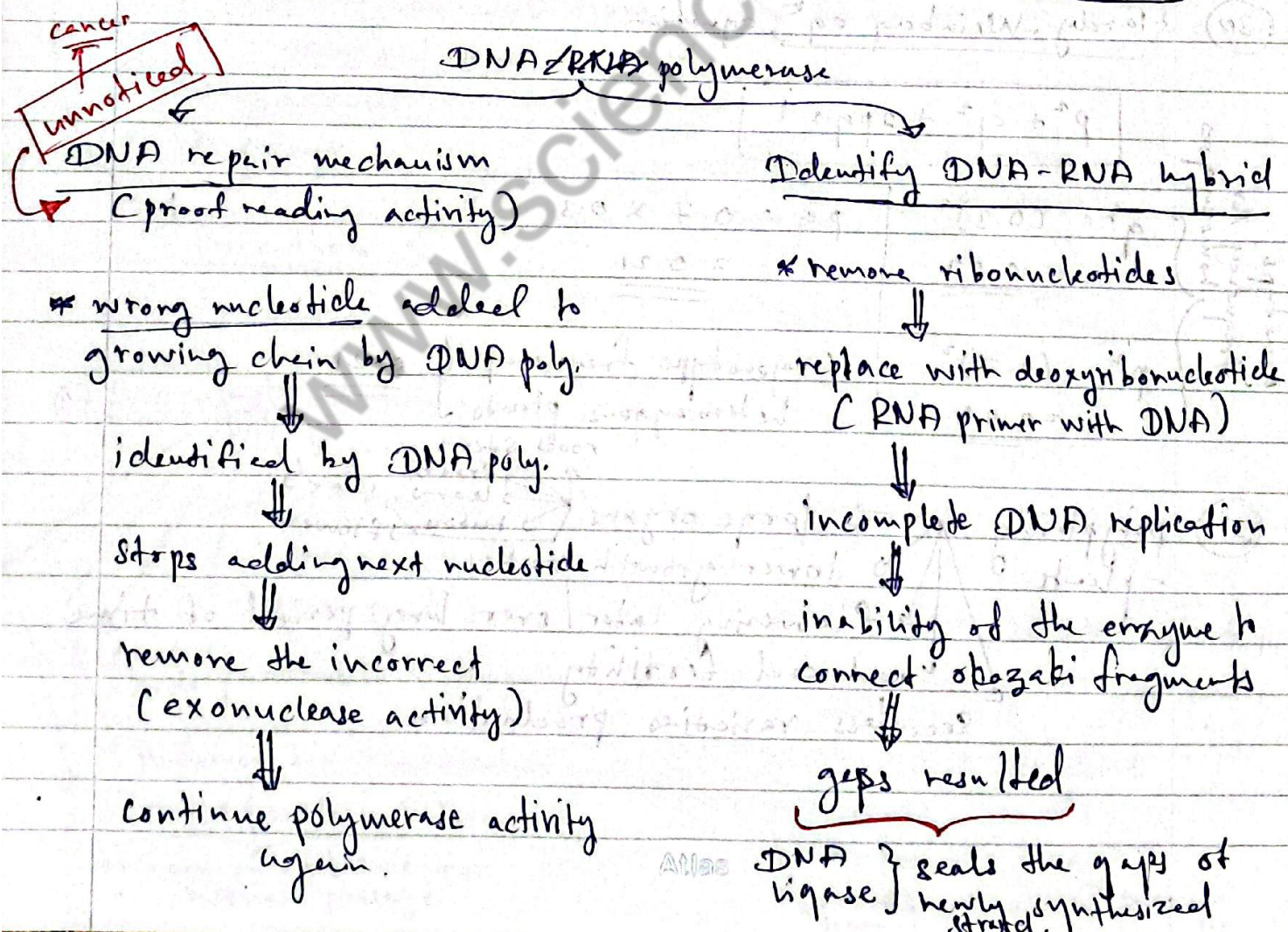
- 1- mRNA moves away for sufficient distance
- 2- binding of 2nd ribo.
- 3- depending the length of mRNA
- 4- no. of ribosomes can be attached mRNA simultaneously
- 5- it is translated actively
- 6- ~~attached~~ ^{assigned} to multiple ribosomes attached to a string of mRNA → polysome / polyribosome
- 7- increases rate of translation
- 8- allow simultaneous translation by several ribosomes.

⑦ Codons

- Start - AUG
- Stop - UAG / UAA / UGA

Cancer
↑
malignant cell
↑
mutation

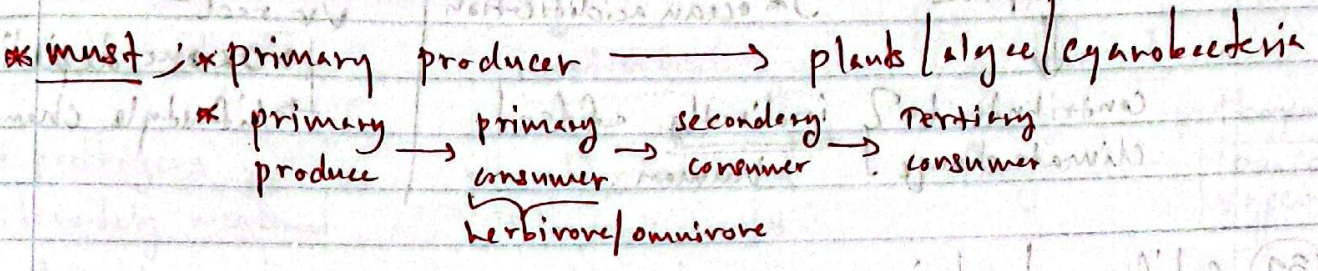
(33) DNA repair mechanism; prevent formation / mutation



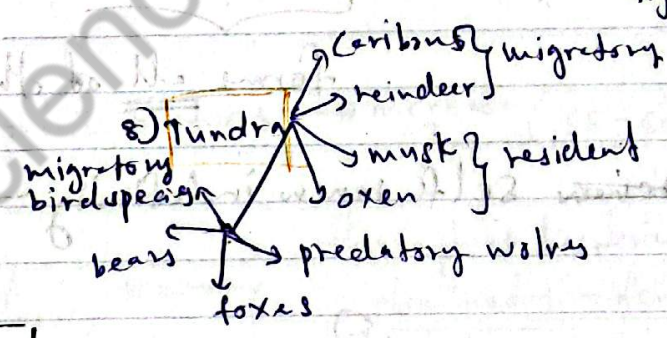
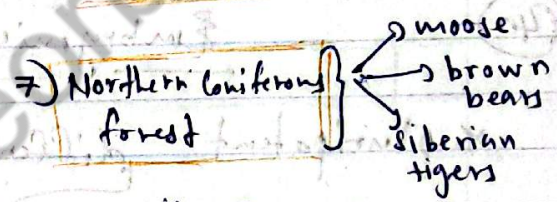
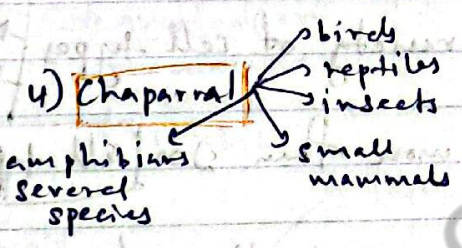
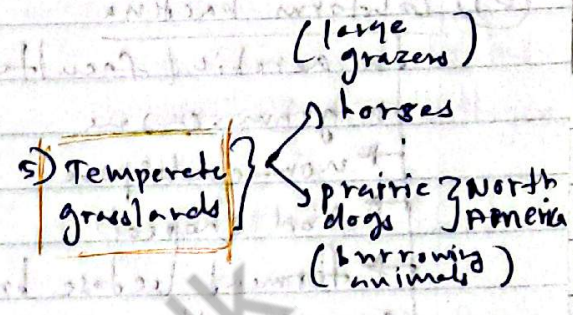
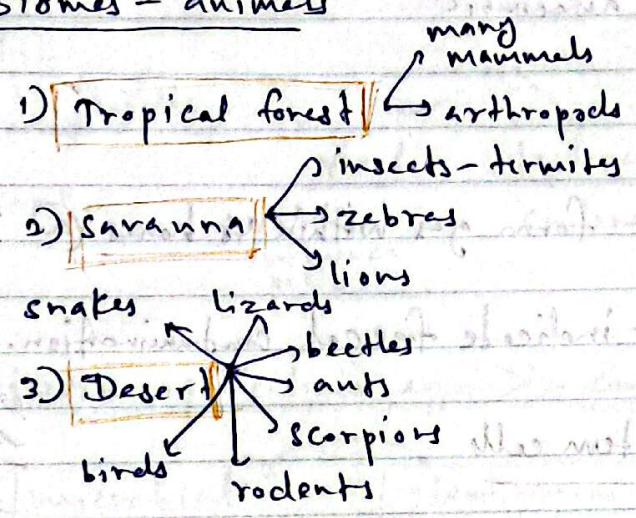
shortest

consumers belong to one ecosystem

Efficient food chain - (natural ecosystem)

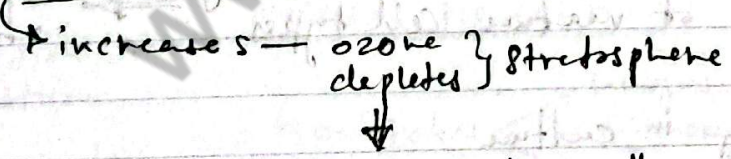


35 Biomes - animals



36 A - ozone depleting substances

CFC / MeBr / Helone / HCFC



UV rad ↑ earth surface

coral reef degradation

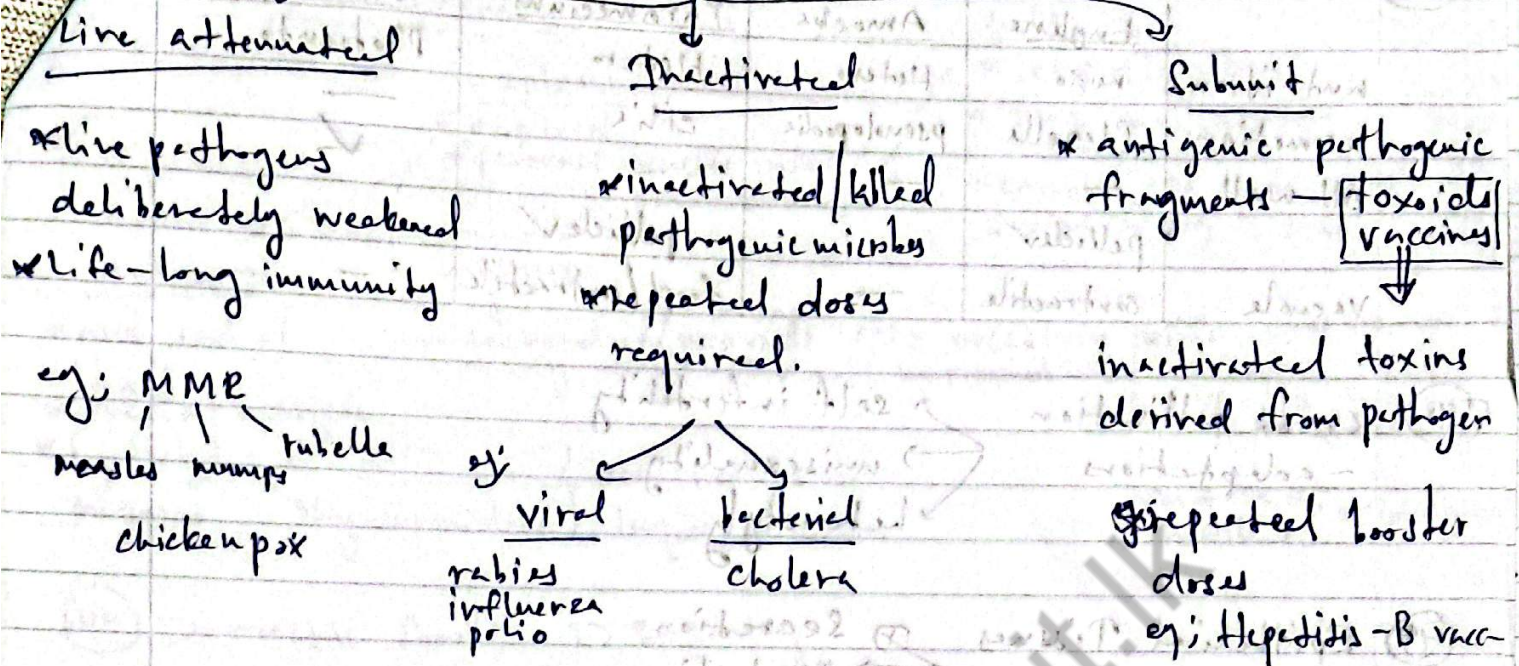
B - CH₄ emitting sources

- anaerobic decomposition
- cattle farming
- partially cultured
- enteric ferment

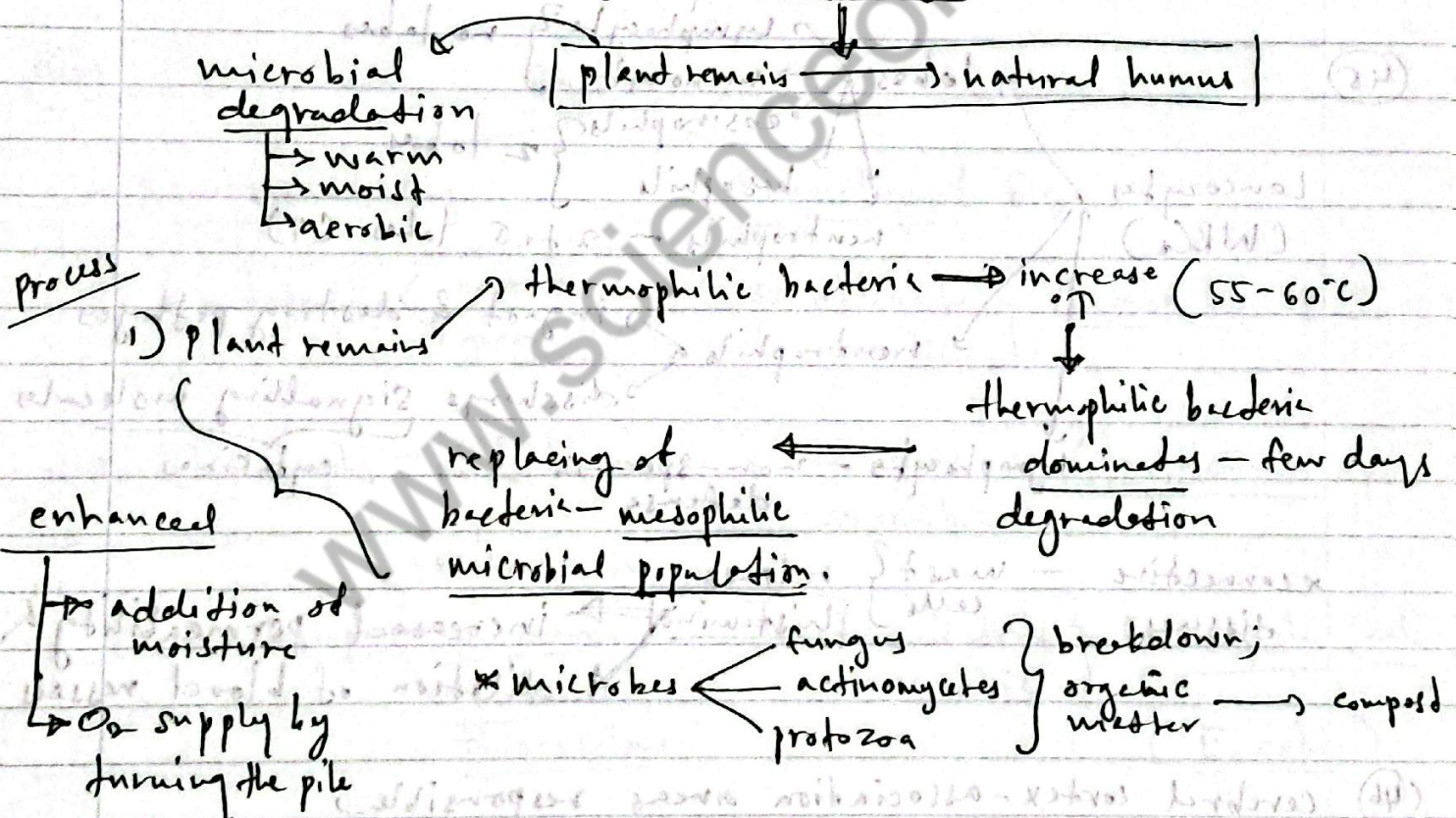
heavy metals

Pb content in drinking water
acid rain

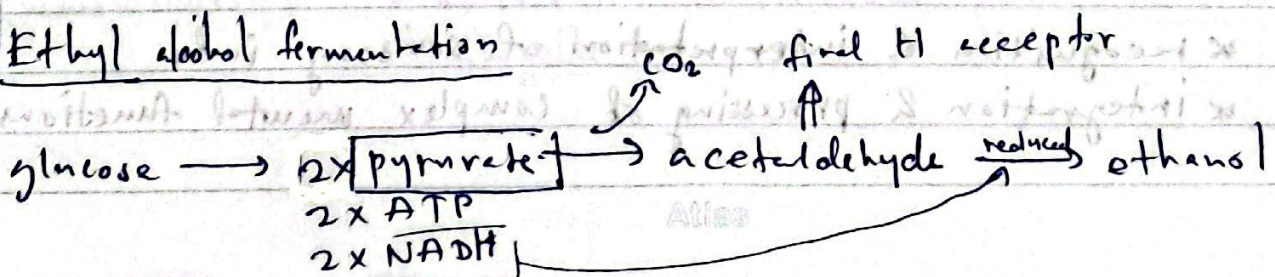
Immunization



(38) Microbes applications agri >> composting



(41) Ethyl alcohol fermentation



(42)

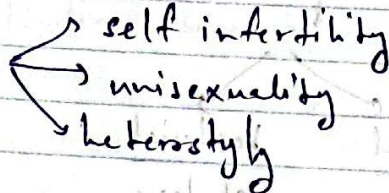
No: _____

Date: ____/____/____

	<u>Euglena</u>	<u>Amoeba</u>	<u>Paramecium</u>	<u>Gracilium</u>
nutrition	mixo	Hetero	Hetero	Photoauto
locomotion	flagella	pseudopodia	cilia	—
cell wall	—	—	—	✓
	pellicle	—	pellicle	—
vacuole	contractile	—	food/contractile	—

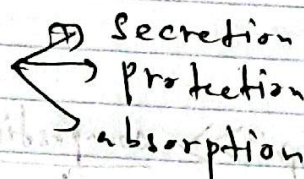
(43)

cross pollination
- adaptations



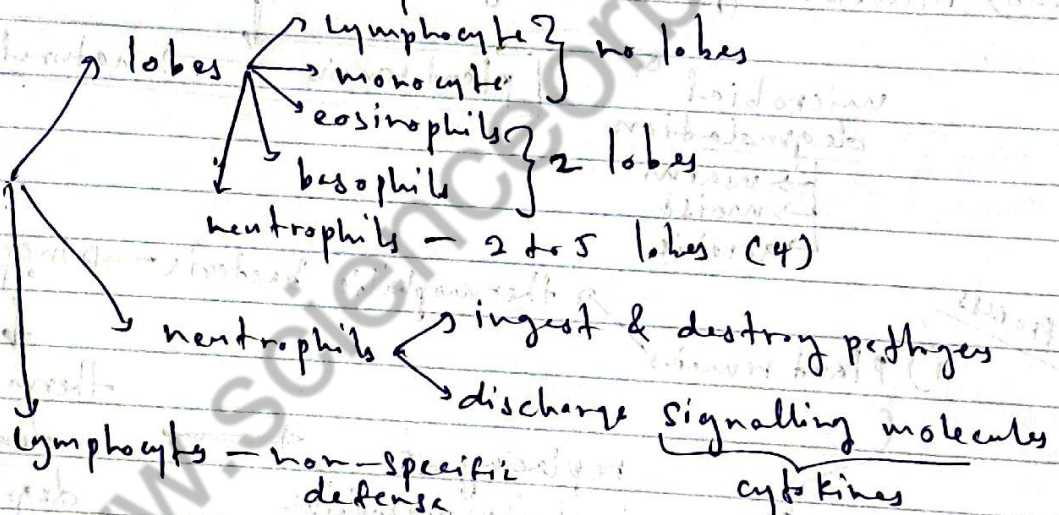
(44)

Epithelial Tissues
- Functions



(45)

Leucocytes
(WBCs)



* connective
tissues

- mast
cells

release

histamine

increased permeability &
dilation of blood vessels

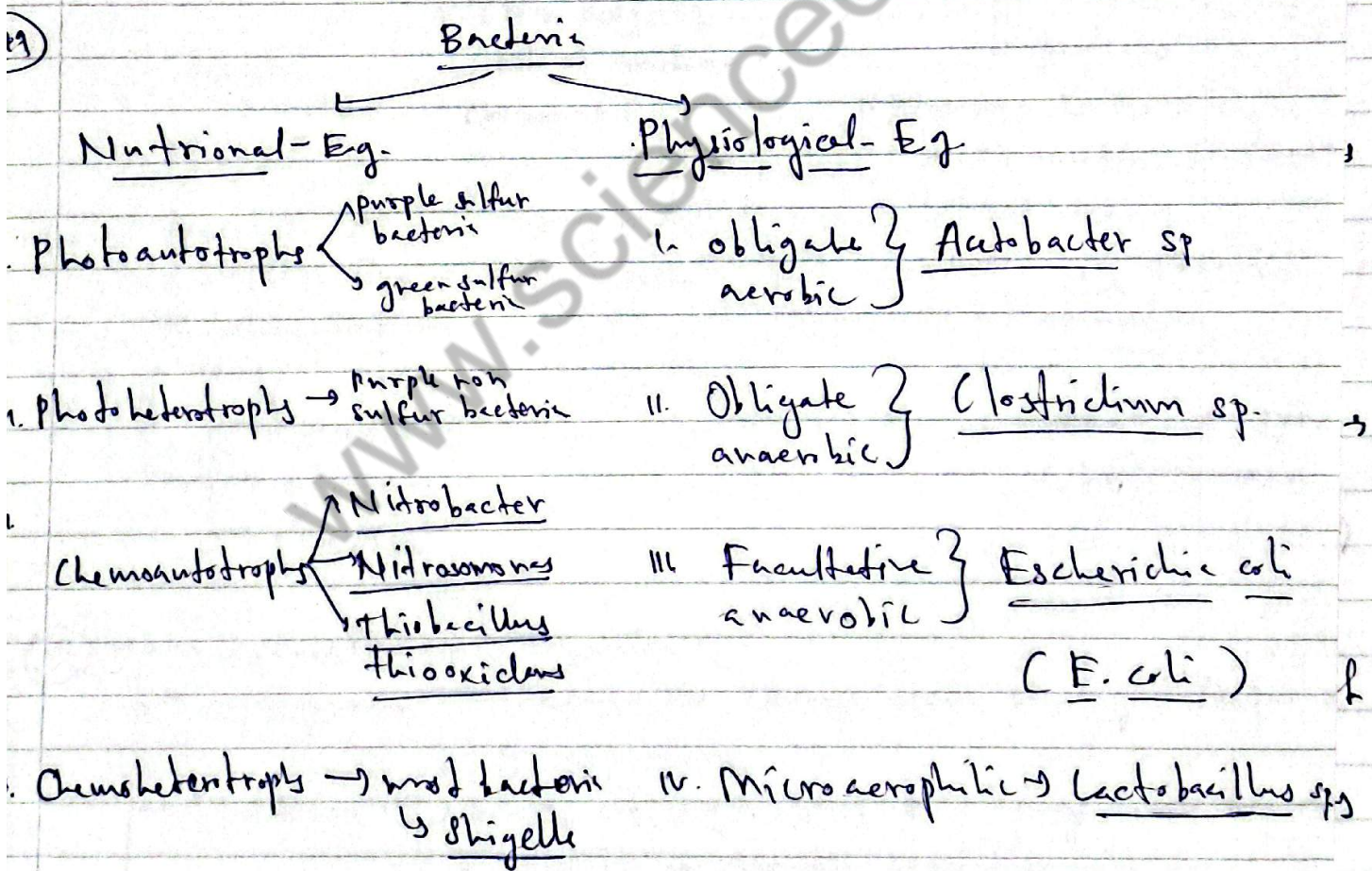
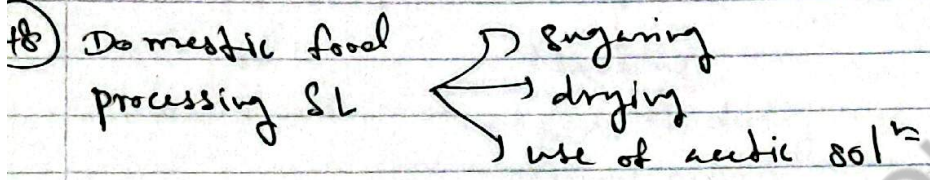
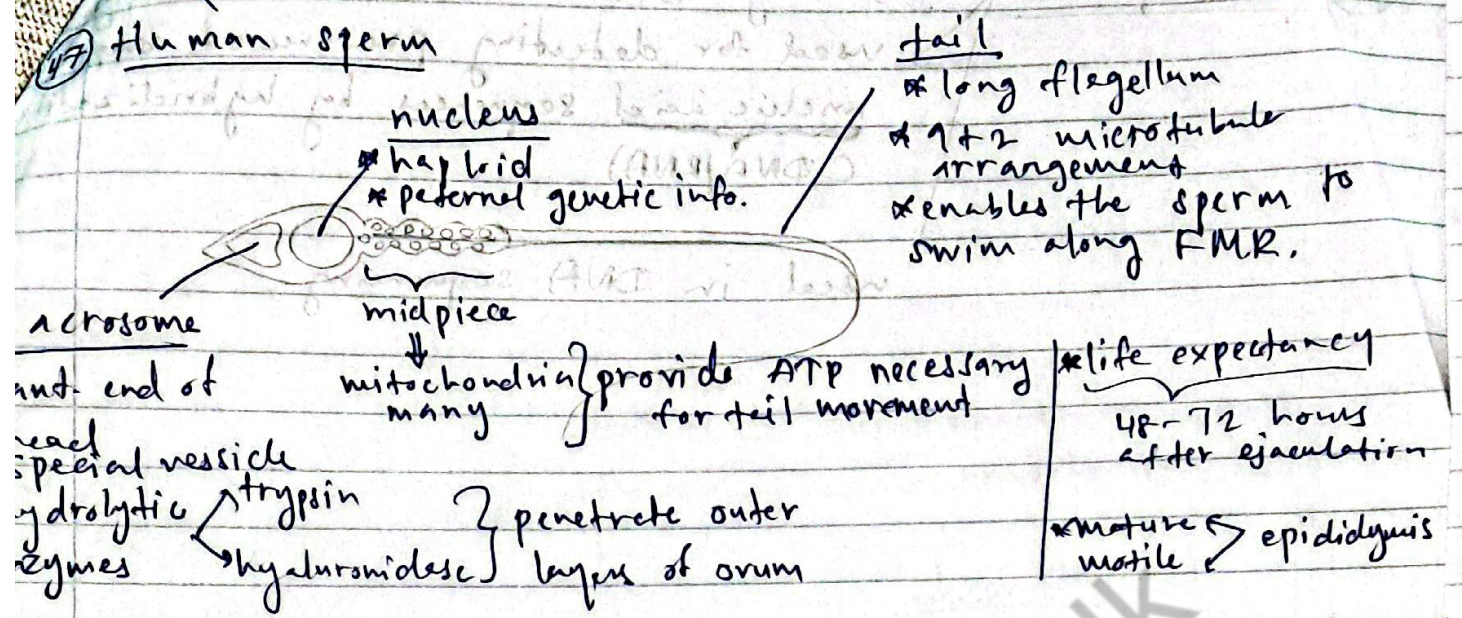
1) Cerebral cortex - association areas responsible;

- * recognition & interpretation of sensory info
- * integration & processing of complex mental functions

not produced before puberty

Date: / /

Human sperm



(15)

DNA probes

single stranded labelled DNA fragments
used for detecting presence of complementary
nucleic acid sequences by hybridization
(DNA/RNA)

used in DNA sequencing.

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